MIL-A-3933/20
25 November 1974

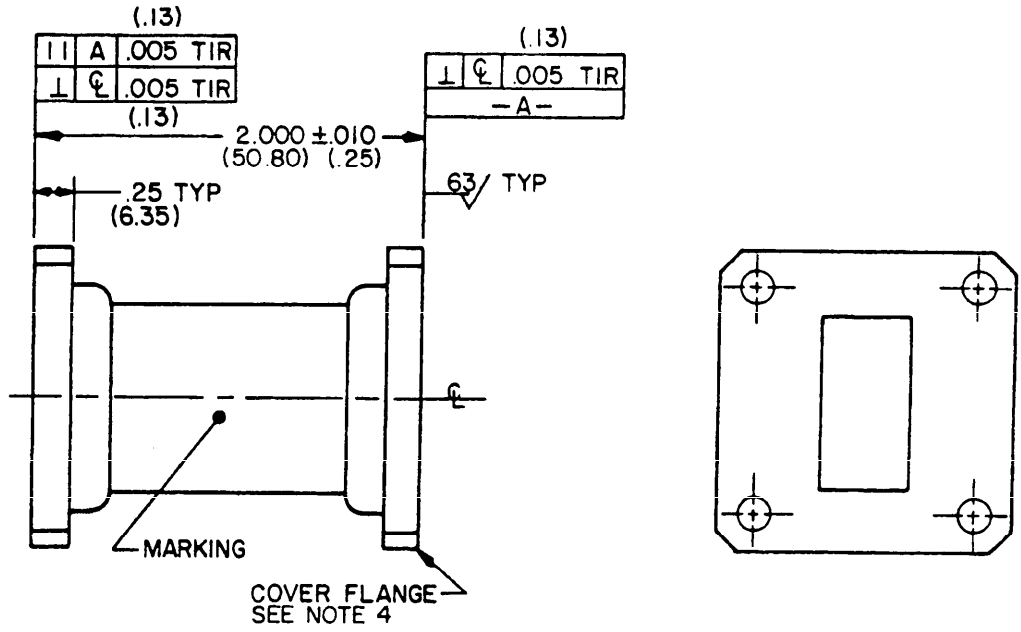
MILITARY SPECIFICATION SHEET

ATTENUATORS, FIXED, WAVEGUIDE

FREQUENCY RANGE: 7 to 18GHZ, CLASS III

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the attenuators described herein shall consist of this document and the latest issue of Specification MIL-A-3933.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Metric equivalents are in parentheses.
4. Use either end as input.

FSC 5985

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ENGINEERING DATA:

Class: III.

Operating frequency range: See table II.

Test frequency f_o for attenuation measurements: See table II.

Operating temperature range: 0°C to 75°C.

Maximum weight: See table I.

REQUIREMENTS:

Dimensions and configuration: See figure.

Attenuation: See table II.

Sensitivity of attenuation: See table II.

Stability of attenuation: See table II.

Power ratings: See table II.

VSWR: See table II.

Flanges: See table I.

Waveguides: See table I.

Pressurization: When pressurized to 20 PSIG air leakage shall not exceed .005 cubic inches per minute, equivalent bubble test is acceptable. 30 PSIG static pressure shall not damage the attenuators.

Quality assurance: Pressurization test shall be an additional group A test. Stability of attenuation after peak power shall be a group B test.

Part number: M3933/20- (and dash number from table I).

TABLE I. Physical characteristics.

Dash number	Flange part number or type	Waveguide part number or type	Weight (oz) Maximum
01	UG-1493/U 1/, 2/	M85/1-158	5
02	M3922/53-012	M85/1-091	4

1/ Except for aluminum alloy and thickness.
2/ Per BuShips dwg RE-D2687618.

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TABLE II. Electrical characteristics

Dash number	Frequency range GHz	Attenuation dB		f_o GHz	Power input		VSWR Maximum over freq range
		Nominal	Deviation at f_o		AV (W) Max	Peak (W) Max	
01	7-11	20.0	±1.0	9.0	1	50	1.25:1
02	11-18	20.0	±1.0	13.5	1	50	1.25:1

Dash number	Sensitivity of attenuation, with max			Stability of attenuation, after max		
	Frequency 1/	Power 2/	Temperature 3/	Temp. cycling	Vibration or shock	Moisture Resistance
01	1.5dB	0.1dB	0.2dB	0.2dB	0.2dB	0.2dB
02	2.0dB	0.1dB	0.2dB	0.2dB	0.2dB	0.2dB

1/ Variation about the value at f_o.
2/ Variation about the value at 50 percent of rated power.
3/ Variation about the value at 25°C.

NOTE: Attenuators covered by this specification are substitutable for the following commercial types:

Part number	Manufacturer's code	Commercial types
M3933/20-01	05869	723343-1
M3933/20-01	14135	XA9-20
M3933/20-02	05869	723343-2
M3933/20-02	14135	KU145-20

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities
Army - EL, MI, SL
Navy - OS
Air Force - 11, 80
DSA - ES

User activities:
Army - MO, SM
Navy - MC, SH, AS
Air Force - 80

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5985-0778-5)

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OMB Approval
No. 22-R255

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NAME OF ORGANIZATION AND ADDRESS

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

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